

### Lesson 9 Set-Theoretic Proofs

In this lesson, we will learn proofs involving sets, specifically how to prove that: (i) a set is a subset of another set, and (ii) two sets are equal. Recall the definition of subset from Lesson 4: set  $A$  is a **subset** of set  $B$  if every element of  $A$  is an element of  $B$ . This is equivalent to: “If  $x \in A$ , then  $x \in B$ .”

- To show that  $A \subseteq B$ , we start with “Let  $x \in A$  (let  $x$  be an arbitrary element of  $A$ ) and go on to show that  $x \in B$ :

$$x \in A \rightarrow \dots \rightarrow x \in B$$

Next,  $A = B$  if they contain exactly the same elements, or  $A \subseteq B$  and  $B \subseteq A$ , so

- To prove that  $A = B$ , we show that  $A \subseteq B$  and  $B \subseteq A$ .

**Proposition 1** If  $A$  is the set of multiples of 4 and  $B$  is the set of even integers, then  $A$  is a subset of  $B$ .

**Proof** Let  $x \in A$ . Then  $x$  is a multiple of 4, and  $x = 4n$  for some integer  $n$ . Hence  $x = 2(2n)$ . Since  $2n$  is an integer,  $x$  is even, and  $x$  is an element of  $B$ .

**Proposition 2** Let  $S = \{2m - 3n : m, n \in \mathbb{Z}\}$ , then  $S = \mathbb{Z}$ .

**Proof** Let  $x \in S$ , then  $x = 2m - 3n$  for some  $m, n \in \mathbb{Z}$ , and  $x \in \mathbb{Z}$ . So  $S \subseteq \mathbb{Z}$ . Next, let  $x \in \mathbb{Z}$ . Since  $x = 2(2x) - 3x$ , and  $2x$  is an integer, we have  $x \in S$ . So,  $\mathbb{Z} \subseteq S$ . We have shown that  $S = \mathbb{Z}$ .

Note: there are many ways to show that  $x \in S$ , such as:  $x = 2(-x) - 3(-x)$ .

We have basic set equivalences, analogous to the logical equivalences in Lesson 3.

- **(Commutative)**  $A \cap B = B \cap A$ ,  $A \cup B = B \cup A$
- **(Associative)**  $(A \cap B) \cap C = A \cap (B \cap C)$ ,  $(A \cup B) \cup C = A \cup (B \cup C)$
- **(Distributive)**  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ ,  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- **(Idempotent)**  $A \cap A = A$ ,  $A \cup A = A$
- **(De Morgan)**  $\overline{A \cap B} = \overline{A} \cup \overline{B}$ ,  $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- **(Intersection is subset of union)**

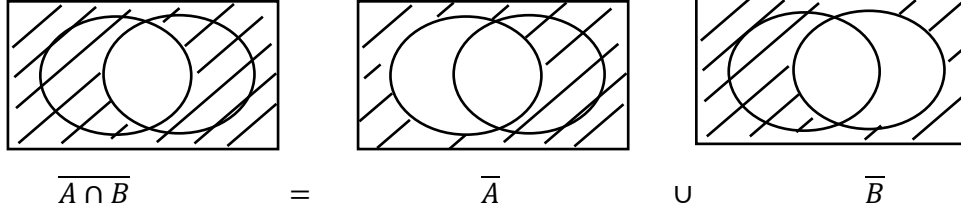
$$A \cap B \subseteq A, \quad A \cap B \subseteq B, \quad A \subseteq A \cup B, \quad B \subseteq A \cup B, \quad A \cap B \subseteq A \cup B$$

We will use the above laws without further proof as they follow directly from the corresponding logical equivalences in Lessons 3. We show some examples.

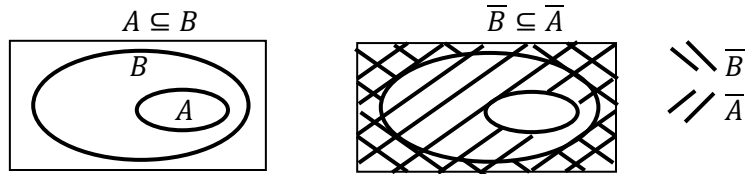
- **(Commutative)** We have  $A \cap B = B \cap A$  since “ $x \in A$  and  $x \in B$ ” if and only if “ $x \in B$  and  $x \in A$ ” ( $p \wedge q = q \wedge p$ ),
- **(De Morgan)** “ $x \in \overline{A \cap B}$ ” if and only if “ $x$  is not both in  $A$  and in  $B$ ”. This is exactly “ $x$  is not in  $A$  or  $x$  is not in  $B$ ” [ $\sim(p \wedge q) = \sim p \vee \sim q$ ], that is,  $x \in \overline{A} \cup \overline{B}$ .

A Venn diagram can be used to effectively *illustrate* a set equivalence, although Venn diagrams are *not* accepted as a formal proof by most, including us.

We give a Venn diagram illustration of the De Morgan's Law  $\overline{A \cap B} = \overline{A} \cup \overline{B}$ .



**Proposition 3** Let  $A$  and  $B$  be sets. If  $A \subseteq B$ , then  $\overline{B} \subseteq \overline{A}$ .



**Proof** Let  $x \in \overline{B}$ , then  $x \notin B$ . We prove that  $x \notin A$ , by contradiction. Suppose  $x \in A$ , since  $A \subseteq B$ ,  $x$  is in  $B$ , contradicting that  $x \notin B$ . Therefore  $x \in \overline{A}$ .

Proposition 3 is the contrapositive equivalence: " $x \in A \rightarrow x \in B$ " = " $x \notin B \rightarrow x \notin A$ ".

**Proposition 4** Let  $A$  and  $B$  be sets. Then  $A \cap B = A$  if and only if  $A \subseteq B$ .

**Proof** ( $\Rightarrow$ ) Suppose  $A \cap B = A$ , then  $A = A \cap B \subseteq B$ .

( $\Leftarrow$ ) Suppose  $A \subseteq B$ , we want to show that  $A \cap B = A$ , i.e.  $(A \cap B) \subseteq A$  and  $A \subseteq (A \cap B)$ . Since  $(A \cap B) \subseteq A$  is true for any sets, we only need to show  $A \subseteq (A \cap B)$ . Let  $x \in A$ . Since  $A \subseteq B$ , we have  $x \in B$ . So,  $x \in A$  and  $x \in B$ , that is,  $x \in A \cap B$ .

**Proposition 5** Let  $A$ ,  $B$ , and  $C$  be sets. If  $A \subseteq B$ , then  $A \times C \subseteq B \times C$ .

**Proof** Let  $(x, y) \in A \times C$ . Then  $x \in A$  and  $y \in C$  by definition of Cartesian product. Since  $A \subseteq B$ ,  $x \in A$  implies that  $x \in B$ . So,  $x \in B$  and  $y \in C$ , and  $(x, y) \in B \times C$ .